Homework 1

Welcome to the first homework. In this homework you will explore descriptive statistics and visualizations of categorical data. Please render this Quarto document and submit the compiled pdf document with your answers to Gradescope *by 11pm on Sunday September 13th*. Please also make sure to mark the correct pages to each question's answers on Gradescope to allow the TAs to grade each problem.

Some functions that could be useful for completing this worksheet are: `length()`, `table()`, `prop.table()`, `get_proportion()`, `barplot()`, and `pie()`. A list of all the functions we have used in class can be found on Canvas. You might also find the following symbols useful: π, $\hat{p}$

For full credit on problems involving R, please be sure to label all of your figures, and to "show your work" by making sure that all values that are answers to questions are printed and visible in your rendered pdf. Typing the name of an object on its own line in an R chunk will print it.

If you need help with any of the homework assignments, please attend the TA office hours which are listed on Canvas and/or ask questions on Ed Discussions forum. Also, please help members of the class by answering questions on Ed Discussions. Finally, if you have conceptual questions, reviewing the class slides and videos will be helpful.

The R chunk below loads the `SDS1000` package, which contains some of the functions we will use in this homework. Please run this chunk once before answering the questions below (you do not need to change anything in it).

```
library(SDS1000)
```

Part 1: Lock 5 questions

Please answer the following questions that are based on questions from the Lock5 textbook. Some symbols that might be useful for answering these questions include $\hat{p}$ and π. You can also write subscripts in LaTeX by using an underscore, e.g., $\beta_{treatment}$.

Exercise 1.1 (6 points): A study by Hamscher et al. (Environmental Health Perspectives, 2003) shows that antibiotics added to animal feed to accelerate growth can become airborne. Some of these drugs can be toxic if inhaled and may increase the evolution of antibiotic-resistant bacteria. The researchers analyzed 20 samples of dust particles from animal farms. Tylosin, an antibiotic used in animal feed that is chemically related to erythromycin, showed up in 16 samples. Based on this study please report:

a. What are the individual cases in this study?

b. What is the variable of interest in this study?

c. What is the statistic of interest in this study, what symbol should be used for this statistic, and what is the value of this statistic? Your choice of statistic and symbol should be one of the statistics/symbol we used in class.

Answers:

a.
b.
c.

Exercise 1.2 (6 points): A study by Valasek and Walter (Pediatric and Adolescent Concussions, Springer, 2012) examined all sports-related concussions reported to emergency rooms for children ages 5 to 18 in the United States over the course of one year. The table below displays the number of concussions in each of the major activity categories.

Activity	Number of concussions
Bicycles	23,405
Football	20,293
Basketball	11,507
Playground	10,414
Soccer	7,667
Baseball	7,433

Activity	Number of concussions
All-Terrain Vehicles	5,220
Skateboarding	4,408
Hockey	4,111
Swimming	3,846
Horseback Riding	2,648
Total	100,952

Please answer the following questions assuming we are interested in concussions in general (i.e., not just concussions for the year of the study):

a. Are these results from a population or a sample? In 1-3 sentences please explain your answer.

b. What proportion of concussions came from playing football? Please write down your answer and denote it with the appropriate symbol (e.g., symbol = value). Make sure the symbol you use also has a subscript to make it clear what the symbol is referring to.

c. What proportion of concussions came from riding bicycles? Please write down your answer and denote it with the appropriate symbol. Make sure the symbol you use also has a subscript to make it clear what the symbol is referring to.

d. Can we conclude that riding bicycles is more dangerous to children in the US than playing football? Why or why not?

Answers:

a.
b.
c.
d.

Exercise 1.3 (6 points): A study by Reinhold et al. (Science, 2009) examined animal behavior and found that rats can quickly learn to play hide-and-seek with humans. The rats seem to love the game, squeaking and jumping for joy when they find the human or are found by the human. The game is played in a large room with multiple boxes. Some of the boxes are transparent and easy to see through, while others are opaque and the inside cannot be seen. Rats go in and out of these boxes and the researchers studied whether they treated the boxes differently depending on whether they were hiding (and didn't want to be seen) or seeking (and didn't care if they were seen). For rats who had been learning the game for about a week, the data on the number of times the rats entered the two types of boxes in each situation is shown in the table below:

Situation	Transparent boxes	Opaque boxes	Total
Hiding	15	38	53
Seeking	17	14	31
Total	32	52	84

Based on this study please report:

a. When the rats were hiding, what proportion of the time did they go into opaque boxes? Please denote your answer with the appropriate symbol, using a subscript to make it clear which group the proportion is referring to.

b. When the rats were seeking, what proportion of the time did they go into opaque boxes? Again, please denote your answer with the appropriate symbol and subscript.

c. Find the difference between the two proportions you calculated in parts (a) and (b). Then, in 1-2 sentences, describe what this difference suggests about whether the rats treated the two types of boxes differently when they were hiding compared to when they were seeking.

Answers:

a.
b.
c.

Part 2: Analyzing data from a survey about the Oxford comma

The Oxford comma is a comma used before the conjunction in a list of three or more items. For example, in the list "apples, oranges, and bananas," the Oxford comma is the comma before the "and." As discussed in the book Eats Shoots & Leaves, Oxford commas can clarify meaning and avoid ambiguity in sentences. However, to paraphrase a song by Vampire Weekend, does anyone really care about the Oxford comma?

To address the question of whether most Americans care about Oxford commas, the website FiveThirtyEight conducted a survey where they asked people "How much, if at all, do you care about the use (or lack thereof) of the serial (or Oxford) comma in grammar?" They also collected other information about the demographics and grammatical preferences of each survey participant. The results were published in an article titled "Elitist, Superfluous, Or Popular? We Polled Americans on the Oxford Comma".

The R chunk below loads the data that was collected in their survey into a data frame called `comma_survey`, for all participants who answered every question on the survey. Each row (case) corresponds to one survey respondent, and the variable we are most interested in is `care_oxford_comma`, which records how much the respondent cares about the Oxford comma. The possible values for this variable are "Not at all", "Not much", "Some", and "A lot".

Let's analyze this data to address the very important question of whether most Americans care about Oxford commas!

```r
# load the survey data (don't change this line)
load("comma_survey_nomissing.Rda")
```

Exercise 2.1 (6 points): To start, please do the following:

a. Extract the column `care_oxford_comma` as a vector from the `comma_survey` data frame (hint: use the $ symbol), and assign this vector to an object called `oxford_care`.
b. Using the `oxford_care` vector, print out how many people were surveyed in this study, and report that number in the answer section.
c. In the answer section, also report what you believe the population of interest is that this survey is trying to learn about.

Hint: to view the data frame you can load the data on the console and then use the `View()` function on the console as well. Remember that you can't use the `View()` function in a Quarto document and also that your Quarto document does not have access to the objects in your R environment. Instead, R can only access objects that are created inside the Quarto document itself.

Answer:

Exercise 2.2 (6 points): Now that you have the `oxford_care` vector, please use the `table()` function on this object to create a count of how many people gave each response, and assign the results to an object called `oxford_comma_counts`. Report which response was given most often.

Answer:

Exercise 2.3 (6 points): Next, please calculate the **relative frequencies** (i.e., the proportions) of how often each response was given, and print out the results to "show your work". Then please answer the following in the answer section:

a. What proportion of people care "A lot" about the Oxford comma, and what proportion of people care "Not at all"?
b. Is the proportion of people who care "A lot" a statistic or a parameter? Which symbol that we used in class should be used to denote it? Please use a subscript to make it clear what the symbol is referring to.

Answers:

a.
b.

Exercise 2.4 (6 points): Now create a bar chart of the counts of how many people gave each response. As always, be sure to label your axes. Also, suppose FiveThirtyEight had conducted another survey 2 years later that asked the same question. In the answer section, report whether you think the proportion of people who care "A lot" about the Oxford comma would be different in this new survey, and why or why not.

Answers:

Exercise 2.5 (6 points): Next create a pie chart of the proportions of how many people gave each response. In the answer section, report whether you think the bar chart or the pie chart is a better way to visualize this data and why?

Bonus (0 points): see if you can figure out how to change the color of the segments of the pie chart.

Answer:

Exercise 2.6 (6 points): Finally, the survey conducted by FiveThirtyEight also collected information about several other variables. Apart from `respondent_id`, the variables in the data frame are: `gender`, `age`, `household_income`, `education`, `location`, `more_grammar_correct`, `heard_oxford_comma`, `care_oxford_comma`, `write_following`, `data_singular_plural`, `care_data`, and `care_proper_grammar`.

You can view the full data frame by using the `View()` function on the console, and you can learn more about the survey by reading the original FiveThirtyEight article (the link is at the top of this part of the homework).

Please explore one of the *other* variables in this data frame by visualizing it and/or by calculating the proportions of people in each of its categories. In the R chunk below, please show your work exploring this other variable, and in the answer section, please report what your analysis shows.

Note: Our grading will be fairly relaxed on this question so do not stress too much about coming up with the perfect analysis here, but do see if you can find something interesting and explain why you think it is interesting.

Answer:

Reflection (6 points)

Please reflect on how the homework went by going to Canvas, going to the Quizzes link, and clicking on Reflection on homework 1

Also, if you have not done so already, please be sure to fill out the class background survey. Three points will be deducted from this homework if the background survey has not been filled out by the time homework 1 is due.